HNDIT 4052

Programming Individual Project

Test Cases

Smart Study Planner with AI Suggestions

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1) Register & login page

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 001	Register with valid details	Kamal kamal@gmail.com 123456	Account created successfully	Account created and stored in MongoDB	Pass
TC 002	Register with empty fields	Empty Name/ Email/ Password	Validation error displayed	Validation error displayed	Pass
TC 003	Register with existing email	Existing email	Registration blocked	Registration blocked	Pass
TC 004	Register with invalid email	abc123	Invalid email error	Error message displayed	Pass
TC 005	Login with valid credentials	kamal@gmail.com 123456	Login successful	Redirected to Dashboard	Pass
TC 006	Login with wrong password	kamal@gmail.com 111111	Login denied	Invalid password message displayed	Pass
TC 007	Login with unregistered email	test@gmail.com	Login denied	User not found message displayed	Pass

2) Subject page

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 008	Add new subject	OOP, hard, high	Subject saved	Subject displayed in table	Pass
TC 009	Add subject with empty field	Subject name empty	Validation error	Error displayed	Pass
TC 010	Edit subject	Change hard → medium	Subject updated	Updated successfully	Pass
TC 011	Delete subject	Click subject	Subject removed	Subject deleted	Pass
TC 012	Add hard subject	ML, hard, high	Subject added	Saved successfully	Pass
TC 013	Add medium subject	ICT, medium, medium	Subject added	Saved successfully	Pass
TC 014	Add easy subject	English, Easy, Low	Subject added	Saved successfully	Pass

3) Task page

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 008	Add new subject	OOP, hard, high	Subject saved	Subject displayed in table	Pass
TC 009	Add subject with empty field	Subject name empty	Validation error	Error displayed	Pass
TC 010	Edit subject	Change hard → medium	Subject updated	Updated successfully	Pass
TC 011	Delete subject	Click subject	Subject removed	Subject deleted	Pass
TC 012	Add hard subject	ML, hard, high	Subject added	Saved successfully	Pass
TC 013	Add medium subject	ICT, medium, medium	Subject added	Saved successfully	Pass
TC 014	Add easy subject	English, Easy, Low	Subject added	Saved successfully	Pass

4) Timetable page

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 022	Save study slot	2026/7/1 10-12	Slot saved	Displayed in Saved Study Times	Pass
TC 023	Save slot with empty fields	Empty date	Validation error	Error displayed	Pass
TC 024	Invalid time range	12-10	Validation error	Error displayed	Pass
TC 025	Display saved slots	Existing slots	Show slots	Displayed correctly	Pass
TC 026	Generate timetable	Subjects + Tasks	Timetable generated	Displayed successfully	Pass
TC 027	Generate timetable with tasks	ML Assignment	Task allocated	Displayed correctly	Pass
TC 028	Generate by difficulty	Hard subject	More time allocated	Generated correctly	Pass
TC 029	Generate without subjects	No subjects	No timetable	Empty result displayed	Pass
TC 030	Generate without tasks	No tasks	Subject-only schedule	Generated successfully	Pass

5) Dashboard page

Test Case ID	description	Input data	Expected outcome	Actual output	Status
TC 031	Today's Targets	Today's task	Display target	Displayed correctly	Pass
TC 032	Upcoming Tasks	Pending tasks	Display upcoming tasks	Displayed correctly	Pass
TC 033	Study Schedule	Timetable data	Display schedule	Displayed correctly	Pass
TC 034	Display schedule	Completed tasks	Show progress %	Displayed correctly	Pass

6) Progress page

Test Case ID	description	Input data	Expected outcome	Actual output	Status
TC 031	Today's Targets	Today's task	Display target	Displayed correctly	Pass
TC 032	Upcoming Tasks	Pending tasks	Display upcoming tasks	Displayed correctly	Pass
TC 033	Study Schedule	Timetable data	Display schedule	Displayed correctly	Pass
TC 034	Display schedule	Completed tasks	Show progress %	Displayed correctly	Pass

7) Suggestions page

Test Case ID	description	Input data	Expected outcome	Actual output	Status
TC 031	Today's Targets	Today's task	Display target	Displayed correctly	Pass
TC 032	Upcoming Tasks	Pending tasks	Display upcoming tasks	Displayed correctly	Pass
TC 033	Study Schedule	Timetable data	Display schedule	Displayed correctly	Pass
TC 034	Display schedule	Completed tasks	Show progress %	Displayed correctly	Pass

8) Notification page

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 048	Display unread notifications	Existing notification records	Notifications displayed	Notifications displayed correctly	Pass
TC 049	Mark notification as read	Click on notification button	Status updated	Notification successfully marked as Read	Pass
TC 050	Upload PDF	Assignment.pdf	PDF uploaded successfully	Displayed successfully	Pass

9) Calendar & settings page

Test Case ID	Description	Input data	Expected outcome	Actual output	Status
TC 042	Calendar deadline display	Task deadline	Show on calendar	Displayed correctly	Pass
TC 043	Update profile	New name	Profile updated	Updated successfully	Pass
TC 044	Change password	Old + New password	Password changed	Updated successfully	Pass
TC 045	Enable dark mode	Click icon	Dark theme enabled	Enabled successfully	Pass
TC 046	Disable dark mode	Click icon	Light theme enabled	Light theme enabled	Pass
TC 047	Logout	Logout button	Session ended	Redirected to home page	Pass